# Harshit Ravi Kumar Ambati

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 8.14 |
| SKVS Junior Kalasala | Class XII (BIEAP) | 2020 | 94.80% |
| Narayana EM High School | Class X (BSEAP) | 2018 | 9.80% |

## SKILLS

1. **HDL & Programming:** Verilog, Python, C (basic)
2. **ASIC / EDA Tools:** Yosys, OpenLane/OpenROAD, Magic, Cadence Virtuoso, Xilinx Vivado, LTspice, Ansys HFSS (basic)
3. **Design Flows :** ASIC Design Flow(Fundamentals), FPGA Design Flow(Fundamentals)

## PROJECTS

- **RTL-to-GDSII ASIC Implementation of a 8-bit Counter:**
- Implemented an 8-bit counter in Verilog HDL and carried out the complete RTL-to-GDSII ASIC design flow using Yosys, OpenLane/OpenROAD, and Magic.
- Performed logic synthesis, floorplanning, placement, clock-tree synthesis, routing, and physical verification to generate the final chip layout.
- Verified the physical design using DRC/LVS checks and generated the final GDSII layout.
- **RISC-V Single-Cycle Processor Design using Verilog HDL:**
- Designed and implemented a single-cycle RISC-V processor using Verilog HDL, developing the datapath and control logic required for instruction execution.
- Implemented key processor components including the Program Counter (PC), instruction memory, register file, ALU, control unit, immediate generation, and data memory.
- Integrated these modules to support the instruction fetch, decode, execute, memory access, and write-back operations within a single clock cycle.
- **Asynchronous FIFO Design using Verilog HDL:**
- Designed and verified a dual-clock asynchronous FIFO for reliable data transfer between independent write and read clock domains.
- Implemented Gray-code-based read/write pointers with two-stage synchronizers to safely transfer pointer information across clock domains.
- Developed full and empty flag generation logic based on synchronized Gray-code pointer comparison to prevent FIFO overflow and underflow.
- **RTL Analysis & Functional Verification of Hybrid PUF-Based Hardware Security IP using Verilog HDL:**
- Studied and analyzed the existing Multiple PUF-CPRNG-based IP authentication RTL architecture, including the PUF, CRP generation, authentication, and User-IP access-control modules.
- Developed and executed Verilog testbenches to functionally verify modules investigated abnormal behavior in the SPUF response generation.
- Identified and analyzed issues such as constant response bits during simulation and examined post-implementation timing simulation results to assist RTL-level debugging.

## RELEVANT COURSES

**Digital:**

Digital IC Design, CMOS Digital Design, Digital IC Design Lab

**Analog:**

Analog IC Design, Analog IC Design Lab

**Devices:**

Memory Device Technology for AI/ML Technology

## THESIS

2026 - Present

**Guide:** Dr. Naresh Kumar Emani

**Nanophotonic Device Characterization:**

- Investigating the optical characterization of plasmonic nanostructures and quantum emitters using Fluorescence Lifetime Imaging Microscopy (FLIM), confocal lifetime microscopy, and Ocean Optics QE Pro-FL spectroscopy.
- Characterizing transmission, emission, and scattering spectra of plasmonic structures and Rhodamine 6G samples, with optimized spectrometer parameters of 500 ms integration time, 8 scans averaged, and boxcar width of 4.
- Obtained a reference intensity of approximately 1.9 × 10 counts with dark counts below 14 counts, and analyzed Rhodamine 6G absorption around 500-560 nm and fluorescence emission around 560-700 nm.
- Characterized gold plasmonic structures exhibiting very low transmitted intensity of approximately 90 counts compared with ~110,000 counts for the reference, with an observed emission range of 540-600 nm, and studied their bright-field/dark-field imaging and scattering behavior, including comparison of single and double plasmonic structures.

## EXTRACURRICULAR

1. **Live-in-Labs:** Rural community engagement for basic amenity improvement
2. **SobhaKruth Nama Ugadi:** Volunteer, crowd management
3. **Student Social Responsibility Project:** Created YouTube podcasts on motivational content
4. **Clean-up Drive:** Participated in hostel-area cleanliness initiative